MILITARY SPECIFICATION

PROJECTILES, 155MM, HE: M692 and M731, NON-METALLIC HARDWARE FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Non-metallic Hardware for Projectiles, 155MM, HE: M692 and M731.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

Federal Test Method Standard No. 601 - Rubber: Sampling and Testing

FSC: 1320

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MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-417	- Rubber, Compositions, Vulcanized General Purpose, Solid, (Symbols and Tests)
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

U.S. ARMY ARMAMENT COMMAND

PICATINNY ARSENAL

9298394	- Pad, Disc
9298332	- Pad, Plate
9298327	- Pad
9298328	- Pad, Shelf Disc
9298329	- Pad, Wedge

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Hardware.-The hardware shall comply with all requirements specified on the drawings referenced in Section 2 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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3.4 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this hardware:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Pad, Disc	9298394	25
Pad, Plate	9298332	25
Pad	9298327	25
Pad, Shelf Disc	9298328	25
Pad, Wedge	9298329	25

4.3.2 Inspections to be Performed.-See MIL-A-48078 and Table I herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48376 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Non-Metallic Hardware				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
	Pad, Disc (Dwg. 9298394)				
	Examination for Defects	25	-	3.2	4.4.2.5
	Durometer Hardness	10	-	3.2	4.5.2
	Material Certification	-	-	3.1	4.5.4
	Pad, Wedge (Dwg. 9298329)				
	Examination for Defects	25	-	3.2	4.4.2.3
	Durometer Hardness	10	-	3.2	4.5.2
	Material Certification	-	-	3.1	4.5.4
	Pad, Plate (Dwg. 9298332)				
	Examination for Defects	25	-	3.2	4.4.2.4
	Material Certification	-	-	3.1	4.5.4
	Pad, (Dwg. 9298327)				
	Examination for Defects	25	-	3.2	4.4.2.1
	Durometer Hardness	10	-	3.2	4.5.2
	Material Certification	-	-	3.1	4.5.4
	Pad, Shelf Disc (Dwg. 9298328)				
	Examination for Defects	25	-	3.2	4.4.2.2
	Material Certification	-	-	3.1	4.5.4

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects, and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48376 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Pad		9298327
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major B					
131	Durometer Hardness	5	5-0-1	3.1	4.5.1
Minor					
201	Thickness		0.65%	3.2	Gage
202	Length*		0.65%	3.2	Gage
203	Width*		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.3	Visual
205	Material Certification			3.1	4.5.3

NOTES:

*See paragraph 4.5.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48376 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298328
4.4.2.2	Pad, Shelf Disc				NEXT HIGHER ASSEMBLY NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major A					
101	Thickness		0.40%	3.2	Gage
102	Diameter*		0.40%	3.2	Gage
103	Diameter of hole*		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.3	Visual
202	Material Certification			3.1	4.5.3

NOTES:
*See paragraph 4.5.2

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48376 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Pad, Wedge		9298329
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major A					
101	Thickness		0.40%	3.2	Gage
Major B					
131	Durometer Hardness	5	5-0-1	3.1	4.5.1
Minor					
201	Radius*		0.65%	3.2	Gage
202	Height*		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.3	Visual
204	Material Certification			3.1	4.5.3

NOTES:
*See paragraph 4.5.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48376 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Pad, Plate				9298332
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major A					
101	Thickness		0.40%	3.2	Gage
Minor					
201	Radius*		0.65%	3.2	Gage
202	Height*		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.3	Visual
204	Material Certification			3.1	4.5.3

NOTES:
*See paragraph 4.5.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48376 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Pad, Disc		9298394
			NEXT HIGHER ASSEMBLY
			NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major A					
101	Thickness		0.40%	3.2	Gage
102	Diameter*		0.40%	3.2	Gage
103	Diameter of hole*		0.40%	3.2	Gage
Major B					
131	Durometer Hardness	5	5-0-1	3.1	4.5.1
Minor					
201	True position of hole with outside diameter*		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.3	Visual
203	Material Certification			3.1	4.5.3

NOTES:
*See paragraph 4.5.2

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests specified, herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078.

4.5 Test Methods and Procedures.-

4.5.1 Durometer Hardness of Rubber Parts.-The parts to be subjected to this test shall be tested as specified in Method 3021 of Federal Test Method Standard No. 601. Five (5) readings shall be taken on each part using the hardness scale specified in MIL-STD-417. The median value of the five readings shall be obtained by arranging the value of the readings from lowest to highest. The median reading will be the third (center) value in the numerically arranged readings. If the median hardness of any part fails to comply with the applicable hardness requirement (See MIL-STD-417) for the grade number the lot represented by the sample parts shall be rejected. (Non-destructive test).

4.5.2 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

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All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.3 Material Certification.-The contractor shall provide certification that each lot of material used to produce each part number complies with all material requirements. The certification for rubber shall include test results or statements which provide specific reference to each applicable requirement of MIL-STD-417 and assurance of compliance with each requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level C.-The hardware shall be preserved and packaged in accordance with MIL-STD-1169. Each package shall contain only one part number. Bulk packaging may be used. Pad, dividers, etc. shall be used to avoid damage to parts.

5.2 Packing

5.2.1 Level C.-The hardware, preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169. Each pack shall contain only one part number.

5.3 Marking.-Marking of the carton shall comply with MIL-STD-116

5.4 Shipping.-When hardware from more than one lot is shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended Use.-The components covered by this specification are intended for use in the M692 and M731 Projectiles.

6.2 Ordering Data.-See MIL-A-48078.

6.3 Submission of Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, Attn: SARPA-QA-T-T, Dover, New Jersey 07801.

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6.4 Submission of results of contractor-conducted examinations and tests.-Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1320-A320

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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